

Fano's Inequality

$$\underbrace{m}_{\text{minimax}} := \inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}(\underbrace{\Phi(\rho(\hat{\theta}, \theta))}_{\text{Hypothesis testing error (about J)}})$$

$$\geq \Phi(\delta) \cdot \left[\inf_{\psi} \mathbb{P}(\underbrace{\psi(z)}_{\neq J}) \right]$$

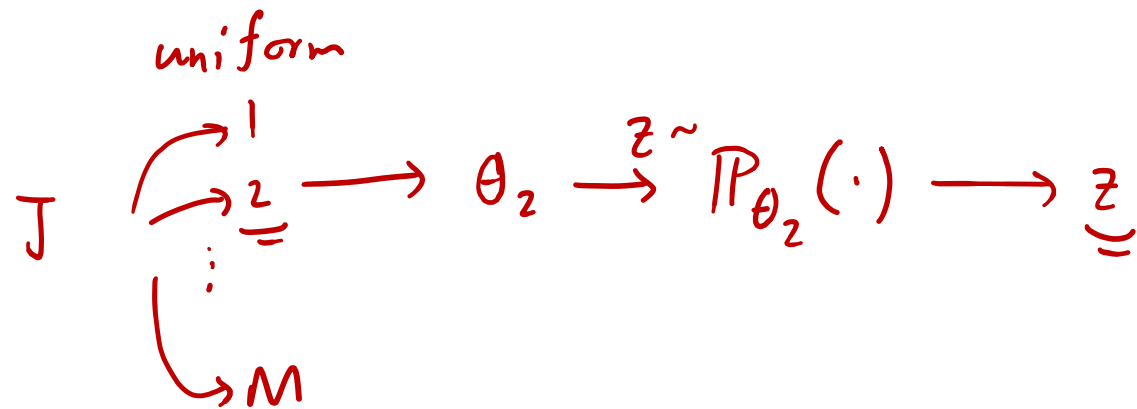
$$\hookrightarrow \{\theta_1, \dots, \theta_m\} \quad 2\delta\text{-separated} \quad \theta_i \in \Theta$$

$$\forall i, j \quad i \neq j \quad \rho(\theta_i, \theta_j) \geq 2\delta$$

$$J \in \{1, \dots, m\}$$

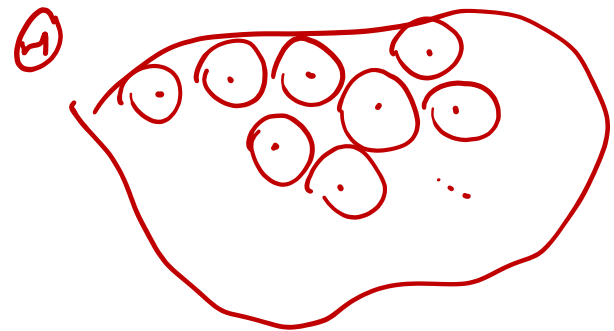
$$\underset{\uparrow}{Z} \sim \mathbb{P}_{\underline{\theta}_J}$$

Fano: $\forall \psi \quad \mathbb{P}(\psi(z) \neq J) \geq 1 - \frac{I(z; J) + \log 2}{\log M} \geq \dots$



Def. $I(z; J) = H(J) - H(J|z)$

Def. $D_{KL}(Q_{z,J} \parallel Q_z \cdot Q_J)$



$$q_e := \mathbb{P}(\overset{\uparrow}{\Psi}(z) \neq J)$$

$$^{(*)} H(q_e) := -\underbrace{q_e}_{\substack{\uparrow \\ q_e}} \log \underbrace{q_e}_{\substack{\uparrow \\ q_e}} - (1-q_e) \log (1-q_e)$$

$$V := \mathbb{I}(\Psi(z) \neq J) \in \{0, 1\}$$

$$\begin{aligned} H(V, J | z) \\ &= \underbrace{H(V | J, z)}_0 + \underbrace{H(J | z)}_{=} \end{aligned}$$

$$\begin{aligned} H(V, J | z) \\ &= H(J | \underset{\uparrow}{V}, z) + \underbrace{H(V | z)}_{\leq H(V)} = H(q_e) \end{aligned}$$

$$\begin{aligned} \textcircled{1} \quad \overset{\text{Shannon}}{H}(X) &= - \sum_{x \in \mathcal{X}} P(X=x) \log P(X=x) \\ H(X|Y) &= - \sum_y \sum_x \underbrace{P(X=x|Y=y)}_{\substack{\uparrow \\ P(Y=y)}} \log P(X=x|Y=y) \end{aligned}$$

$$H(J|V, z) = \underbrace{P(V=0)}_0 \cdot H(J|V=0, z) + \underbrace{P(V=1)}_{q_e} \cdot \underbrace{H(J|V=1, z)}_{\leq \log(M-1)}$$

$$\begin{aligned} H(V, J|z) &= H(J|z) \leq H(q_e) + q_e \log(M-1) \\ &\leq H(q_e) + q_e \log M \end{aligned}$$

$$I(J; z) = H(J) - \underbrace{H(J|z)}$$

$$\Rightarrow H(J|z) = H(J) - I(J; z)$$

$$\underbrace{H(J)}_{\leq \log M} - I(J; z) \leq \underbrace{H(q_e)}_{\leq \log 2} + q_e \log M \leq \log 2 + q_e \log M$$

$$\Rightarrow q_c \geq \frac{\log M}{\log M} - \frac{I(J; z) + \log 2}{\log M}$$

$$= 1 - \frac{I(J; z) + \log 2}{\log M}$$

□

$$n \geq \underbrace{\Phi(\delta)}_{\frac{1}{2}} \cdot \left[1 - \frac{I(z; J)^{\downarrow} + \log 2}{\log M^{\uparrow}} \right]$$

$$\delta \rightarrow 0^+$$

(Z, J)

$$Q_z = \frac{1}{M} \sum_{j=1}^M \underbrace{P_{\theta_j}}_{Q_{z|J}}(z)$$

$$Q_z = \sum_{j=1}^M \underbrace{Q_{z,J}(z, j)}_{Q_{z|J} \cdot \underbrace{Q_J}_{1/M}}$$

$$I(J; z) = D_{KL} (Q_{z,J} \parallel Q_z, Q_J)$$

$$= \sum_z \sum_j Q_{z,J} \cdot \log \frac{Q_{z,J}}{Q_z \cdot Q_J}$$

$$= \frac{1}{M} \sum_z \sum_j P_{\theta_j} \log \frac{\frac{1}{M} \cdot P_{\theta_j}}{\underbrace{\frac{1}{M} \sum_{i=1}^M P_{\theta_i}}_{Q_z} \cdot \frac{1}{M}}$$

$$\overset{J, z, \theta_j}{D_{KL}} (P \parallel q)$$

$$= \sum_x P(x) \log \frac{P(x)}{q(x)}$$

$$= \frac{1}{M} \sum_j \sum_z P_{\theta_j} \log \frac{P_{\theta_j}}{Q_z} \quad \underbrace{\hspace{1cm}}_{D_{KL}(P_{\theta_j} \parallel Q_z)}$$

$$\Rightarrow I(J; z) = \frac{1}{M} \sum_{j=1}^M D_{KL}(P_{\theta_j} \| Q_z) \leftarrow$$

$$\leq \frac{1}{M^2} \sum_{i,j=1}^M D_{KL}(P_{\theta_j} \| P_{\theta_i}) \leftarrow$$

E.g. $\mathcal{N}(\theta, \sigma^2)$ find minimax lower bound on $\hat{\theta}$.

$\{-2\delta, 0, 2\delta\}$ be the 2δ -sep. set.

$$I(J; z) \leq n \cdot \frac{(\theta_i - \theta_j)^2}{2\sigma^2} \leq \frac{2n\delta^2}{\sigma^2} \quad \delta^2 = \underbrace{\frac{\sigma^2}{2n} \cdot \frac{1}{20}}$$

$$= \frac{2n}{\cancel{\sigma^2}} \cdot \frac{\cancel{\sigma^2}}{2n} \cdot \frac{1}{20} = \frac{1}{20}$$

$$\mathbb{P}(\Psi(z) \neq J) \geq 1 - \frac{\frac{1}{20} + \log 2}{\log 3} \geq \frac{1}{4}$$

$$m \geq \Phi(\delta) \cdot \frac{1}{4} = \frac{\sigma^2}{2n} \cdot \frac{1}{20} \cdot \frac{1}{4} = \frac{\sigma^2}{\underbrace{2 \cdot 80n}}_1$$